Evaluating XAI Metrics in Medical AI:
A Cross-Study Co-occurrence Analysis and Evaluation Taxonomy

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Abstract

Explainable AI (XAI) evaluation in medical AI remains fragmented, with limited cross-domain comparison and no consensus on best practices. We conduct a PRISMA-compliant systematic review of 327 records, identifying 52 eligible studies across five clinical sub-domains. Faithfulness dominates current evaluation practice (47/52; 90.4%), while Sufficiency is rarely used (4/52; 7.7%).

Correlation analysis of perturbation-based metrics indicates partial redundancy (AOPC/MoRF vs. Pixel Flipping: $\rho = 0.51$, $p < 0.001$). Mapping studies to a three-layer taxonomy reveals an imbalance between mathematical faithfulness (90.4%), system robustness (34.6%), and human-centred trust (13.5%).

These results highlight a strong emphasis on technical correctness, with limited attention to robustness and clinical trust, suggesting the need for more balanced evaluation frameworks for real-world deployment.

1 Introduction

Explainable Artificial Intelligence (XAI) encompasses a set of methods designed to render machine learning models interpretable by revealing internal mechanisms and the reasoning processes underlying their decisions [1]. In clinical medicine, interpretability is not merely desirable but increasingly mandated: clinical decision-making carries direct consequences for patient safety, and regulatory frameworks across multiple jurisdictions now require that AI-assisted diagnoses be accompanied by interpretable justifications [2]. Despite this regulatory momentum, the metrics used to evaluate explanation quality in medical AI remain poorly standardised. Current evaluation methodologies rely heavily on subjective user studies which, while valuable for assessing clinician perception, are susceptible to confirmation bias and frequently fail to assess the technical fidelity of the generated explanations [3].

This standardisation deficit is compounded by the prevalence of post-hoc explanation methods in clinical AI. Because such explanations are generated after model training, they may not faithfully reflect the model's actual decision-making logic. In medical contexts, this gap is particularly consequential: a persuasive but unfaithful explanation could reinforce a clinician's incorrect judgement rather than correct it, potentially compromising patient outcomes.

The present study focuses on the medical domain and addresses three research questions: (RQ1) Which quantitative evaluation metrics are currently employed to assess XAI explanation quality in medical AI? (RQ2) How do these metrics co-occur across studies, and which pairs exhibit potential redundancy? (RQ3) Where do systematic coverage gaps persist across medical sub-domains? By analysing a focused corpus of 52 peer-reviewed medical XAI studies spanning five sub-domains—oncology, cardiology, neuroimaging, radiology, and clinical decision support—we quantify metric co-occurrence patterns, identify redundancies in current practice, and propose a minimal but comprehensive evaluation framework grounded in a three-layer taxonomy of mathematical faithfulness, system robustness, and human-centred trust.

The principal contributions of this paper are threefold:

1. A systematic cross-study co-occurrence analysis quantifying the distribution, pairing patterns, and correlations of XAI evaluation metrics across five medical sub-domains.
2. A three-layer evaluation taxonomy—mathematical faithfulness, system robustness, and human-centred trust—grounded in empirical usage patterns and validated against the 52-study corpus.
3. A minimal evaluation framework specifying one metric per evaluative dimension, designed to ensure comprehensive coverage without imposing excessive methodological burden.

Figure 1 illustrates the three-layer taxonomy structure and the empirical coverage imbalance motivating this study.

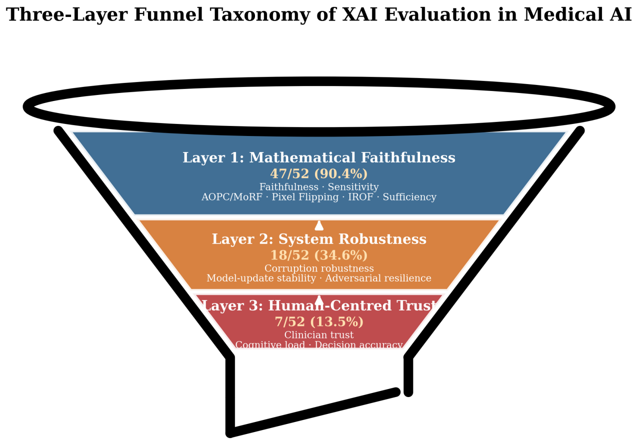

Figure 1: Three-layer funnel taxonomy of XAI evaluation in medical AI. Width represents corpus coverage: Layer 1 (Mathematical Faithfulness, 90.4%), Layer 2 (System Robustness, 34.6%), and Layer 3 (Human-Centred Trust, 13.5%). The narrowing funnel highlights the concentration of evaluation effort in technical fidelity metrics.

2 Background

Artificial intelligence is being rapidly integrated into clinical workflows across diagnostic imaging, prognostic modelling, and therapeutic decision support. However, because complex AI models frequently operate as opaque systems, clinicians cannot readily ascertain how diagnostic or prognostic outputs are generated. This opacity creates a pressing need for XAI methods that render medical AI systems transparent and interpretable [4]. XAI has been applied to domains including clinical decision support [5], breast cancer diagnosis [6], cardiovascular risk prediction [7], and responsible cardiovascular disease screening [8].

A central distinction in XAI is between models that are interpretable by design (intrinsically interpretable models) and methods that generate explanations after training (post-hoc methods). Post-hoc approaches are appealing because they can be applied to virtually any deployed clinical system without requiring architectural modification. However, this flexibility entails a fundamental limitation: there is no guarantee that the explanation accurately represents the model's internal computations, and this fidelity gap remains one of the core unresolved challenges in medical XAI [9, 10].

Two broad paradigms have emerged for evaluating explanation quality in medicine. *Technical evaluation* employs quantitative metrics—such as faithfulness, sensitivity, robustness, and consistency—to assess how accurately an explanation approximates actual model behaviour. *Human-centred evaluation*, by contrast, focuses on the end user, measuring constructs such as clinician trust, cognitive load, and practical usefulness through observational studies and behavioural experiments [11].

While both paradigms have matured considerably, they remain largely disconnected in medical practice. A method that achieves high scores on technical metrics may produce explanations that are opaque or misleading to clinicians, while an explanation that non-specialists find intuitive may lack mathematical fidelity to the underlying model. This disconnect impedes the establishment of shared evaluation standards and limits the comparability of results across medical sub-domains—a problem this paper directly addresses through cross-domain co-occurrence analysis and structured taxonomy construction.

3 Related Work

Efforts to taxonomise XAI methods have produced a substantial but fragmented body of work. Arrieta et al. [1] established the foundational distinction between interpretable-by-design models and post-hoc explanations. Schwalbe and Finzel [12] synthesised over fifty surveys, showing that properties such as faithfulness and stability are widely referenced but inconsistently operationalised. Speith [13] traced this inconsistency to a structural source: mechanism-based versus output-based taxonomies classify the same methods differently.

Medical XAI taxonomies address this gap but remain incomplete. Jin et al. [2] distinguish technical faithfulness from clinical usefulness; Holzinger et al. [9] identify robustness under distributional shift as unresolved for clinical deployment; Tjoa and Guan [4] survey XAI methods broadly without formalising evaluation criteria. Varghese et al. [14] rank metrics using Borda count and multiple correlation techniques, identifying faithfulness and sensitivity as primary axes without cross-domain validation. Reviews in radiology [15–17] confirm that evaluation metrics are inconsistently reported and rarely compared. Reyes et al. [18] argue that explanations must align with clinical reasoning—a requirement frequently absent from benchmark-driven evaluations.

The metrics literature exposes practical consequences of this fragmentation. Rosenfeld [19] shows that most evaluations prioritise algorithmic fidelity while neglecting human-centred dimensions. Kadir et al. [20] confirm that sensitivity, faithfulness, and robustness dominate with limited attention to human factors. Phuong et al. [21] and Klein et al. [22] provide evidence that without structured taxonomies, metric selection remains ad hoc. Recent work on faithfulness under distribution shift—notably the F-Fidelity framework [23]—demonstrates that standard faithfulness metrics may be unreliable when evaluated outside the training distribution, directly motivating the robustness layer in our taxonomy. Complementary analyses [3,24,25] reinforce that the absence of shared evaluation vocabulary impedes meaningful synthesis.

Human-centred evaluation has received increasing attention. Chen et al. [26] identify the systematic absence of human-centred components in medical XAI pipelines. Bauer and Michalowski [27] highlight cognitive load, trust calibration, and decision accuracy as key constructs requiring operationalisation. Dey et al. [28] document continued methodological fragmentation, and Müller et al. [29] emphasise that regulatory demands extend beyond standard faithfulness metrics. Recent surveys [30–32] confirm that standardised frameworks remain underdeveloped, with Ahmed et al. [32] identifying metric standardisation as a major open challenge. Qureshi et al. [33] provide metric-driven assessment of five XAI methods for healthcare tabular models, revealing normalisation inconsistencies that further motivate structured evaluation.

No existing framework jointly spans oncology, cardiology, neuroimaging, radiology, and clinical decision support while integrating mathematical faithfulness, system robustness, and human-centred trust. The present study addresses this gap through systematic co-occurrence analysis and empirically validated taxonomy construction.

4 Methodology

To systematically characterise how XAI evaluation metrics are employed within medical AI, we adopt a four-phase methodology: corpus collection, criterion extraction and normalisation, cross-domain correlation analysis, and taxonomy construction.

4.1 Phase 1: Systematic Corpus Collection

Databases and Search Strategy. We searched PubMed, Scopus, IEEE Xplore, and Google Scholar using query strings combining XAI/explainability terms with evaluation/metrics terms and medical domain terms (e.g., `("explainable AI" OR "XAI") AND ("evaluation" OR "metrics") AND ("medical imaging" OR "clinical")`). Results were restricted primarily to 2020–2025, with seminal earlier works retained. Searches were last updated in January 2026.

Inclusion and Exclusion Criteria. Studies were included if they: (I1) were peer-reviewed in a Q2+ venue or recognised conference (IEEE, ACM, IJCAI, MICCAI); (I2) addressed at least one of five medical sub-domains; and (I3) contained explicit evaluation of XAI quality using at least one quantitative metric or structured human-centred assessment. Studies were excluded if they addressed non-medical AI exclusively, proposed XAI methods without quantitative evaluation, were duplicates, or were published below Q3 without novel domain-specific methodologies.

Screening and Selection. Two independent reviewers screened 327 candidate records. After removing 89 duplicates, 238 records were screened by title and abstract (143 excluded). The remaining 95 full-text articles were assessed against inclusion/exclusion criteria; 43 were excluded (no quantitative XAI metrics: $n = 28$; outside sub-domains: $n = 11$; below venue threshold: $n = 4$). Inter-rater agreement was substantial (Cohen's $\kappa = 0.82$). The final corpus comprises 52 studies, following PRISMA conventions [34].

Corpus Justification. The corpus spans five medical sub-domains with representation from 28 unique venues ranging from top-tier journals (Nature Machine Intelligence, Medical Image Analysis, IEEE TMI) to specialised clinical outlets, ensuring that observed patterns are not artefacts of a single community's conventions. For binary co-occurrence analysis involving six metrics, 52 studies yield adequate statistical power to detect moderate-to-strong associations ($\rho \geq 0.35$) at $\alpha = 0.05$. The corpus incorporates clinician-grounded studies [35–39], imaging-focused analyses [10,40–43], neuroimaging studies [44–46], and human-centred evaluation studies [47,48].

4.2 Phase 2: Criterion Extraction and Normalisation

From each study, we recorded every evaluation criterion and its operationalisation. The same concept frequently appears under different names—"faithfulness," "fidelity," and "completeness" are often used interchangeably [13]. Each criterion is mapped to a canonical label grounded in established literature [19,20], then classified along two dimensions: quantitative versus user-study-based, and overall model behaviour versus individual predictions. Varghese et al. [14] further informed our normalisation by applying Borda count and correlation techniques. The resulting normalised list comprises six metrics—Faithfulness, Sensitivity, AOPC/MoRF, Pixel Flipping, IROF, and Sufficiency—comparable across sub-domains regardless of original terminology.

4.3 Formal Definitions of Evaluation Metrics

We formally define each metric following established conventions [49–51].

Faithfulness measures the correlation between attributed importance and actual feature contribution:

$$\mathrm{Faith}(f, x, \phi) = \mathrm{corr}\big(\phi(x)_i,\ f(x) - f(x_{\setminus i})\big)_{i=1}^{d} \quad (1)$$

Sensitivity captures the maximum explanation change within a perturbation radius ϵ:

$$\mathrm{Sens}_{\max}(f, x, \phi) = \max_{\|x'-x\|\leq\epsilon} \|\phi(x') - \phi(x)\| \quad (2)$$

AOPC/MoRF quantifies the average prediction drop under Most-Relevant-First feature removal:

$$\mathrm{AOPC} = \frac{1}{K+1} \sum_{k=0}^{K} \left\langle f(x^{(0)}) - f\left(x_k^{(\mathrm{MoRF})}\right) \right\rangle_x \quad (3)$$

Pixel Flipping measures prediction drop after flipping the top-k saliency-ranked pixels:

$$\mathrm{PF}(k) = f(x^{(0)}) - f(\tilde{x}_k) \quad (4)$$

IROF averages prediction drops over the removal of semantically coherent segments:

$$\mathrm{IROF} = \frac{1}{S} \sum_{s=1}^{S} \big[f(x) - f(x_{\setminus s})\big] \quad (5)$$

Sufficiency assesses whether highlighted features alone reproduce the prediction (near-zero = sufficient):

$$\mathrm{Suff}(f, x, \phi) = f(x) - f(x_\phi) \quad (6)$$

These six metrics provide complementary perspectives on explanation quality across different evaluation dimensions.

AOPC and Pixel Flipping are mechanistically related—both instantiate perturbation-and-measure—but operate at different granularities (feature-level vs. pixel-level). IROF extends this to semantically meaningful segments, making it particularly suited to medical imaging where anatomical structures correspond to contiguous regions [51]. Sufficiency is conceptually distinct: it assesses whether highlighted features alone sustain the prediction, a property of particular clinical relevance for identifying decision-sufficient evidence sets.

4.4 Phase 3: Cross-Domain Alignment and Correlation Analysis

The six normalised metrics were compared across sub-domains in a comparison matrix. Criteria appearing in $\geq$ 3 sub-domains are classified as *cross-domain*; those in 1–2 as *domain-specific*, reflecting Schwalbe and Finzel's [12] observation that properties such as faithfulness recur broadly but with divergent operationalisations.

Statistical Approach. We compute **Spearman's rank correlation (ρ)** across all 15 pairwise metric combinations using the full 52-study corpus. Each metric is encoded as a binary variable (presence = 1, absence = 0); in this encoding, Spearman's ρ is mathematically equivalent to the phi coefficient (ϕ). Two limitations apply: binary encoding discards information about intensity of metric application, and high co-occurrence does not establish construct equivalence. These are addressed in Section 6.5.

4.5 Phase 4: Taxonomy Construction

The final phase organises aligned criteria into a three-layer structure, building on Kadir et al. [20] while adding an explicit robustness dimension that prior taxonomies have addressed inconsistently [2, 13]. The distinction between technical fidelity (Layer 1) and user-facing evaluation (Layer 3) is well established [2, 20]. Layer 2 is motivated by Holzinger et al. [9] identifying robustness under distributional shift as unresolved, and by our corpus analysis revealing robustness-oriented evaluation as a coherent cluster. Recent work on F-Fidelity [23] further validates this separation by demonstrating that standard faithfulness metrics degrade under out-of-distribution conditions.

Layer 1 (Mathematical Faithfulness) covers whether the explanation accurately reflects model computation, including faithfulness correlation, sufficiency, sensitivity, and fidelity under perturbation.

Layer 2 (System Robustness) addresses whether explanations remain stable when patient inputs shift, models are updated, or adversarial conditions arise. Repetto et al. [41] provide direct evidence that XAI explanations degrade under both natural and adversarial data corruption. The CACTUS framework [52] further demonstrates that explanation robustness under incomplete or missing data requires dedicated evaluation protocols.

Layer 3 (Human-Centred Trust) covers whether explanations help clinicians understand, decide, and calibrate confidence. In clinical contexts, a technically faithful but opaque explanation provides limited practical value [13]. Neuro-symbolic approaches such as XAI-MeD [53] illustrate emerging methods that bridge technical faithfulness with domain-specific interpretability for clinical end users.

Cross-layer criteria (e.g., contrastive explanations) are flagged to maintain transparency about inter-layer connections. The resulting framework functions as both a descriptive map of current evaluation practice and a diagnostic tool for identifying coverage gaps. Figure 1 illustrates the three-layer funnel structure.

5 Results

The following sections report findings from the four-phase methodology, organised by corpus composition, metric co-occurrence, cross-domain alignment, and taxonomy validation.

5.1 Corpus Composition and Metric Distribution

The corpus spans 52 evaluation studies across five sub-domains: oncology ($n = 8$), radiology ($n = 12$), cardiology ($n = 9$), neuroimaging ($n = 7$), and clinical decision support ($n = 16$).[1] Table 1 provides a structured summary.

Six quantitative evaluation metrics were identified. Faithfulness was the most frequently reported (47/52; 90.4%), followed by Sensitivity (34/52; 65.4%), Pixel Flipping (14/52; 26.9%), AOPC/MoRF (13/52; 25.0%), and IROF and Sufficiency (each 4/52; 7.7%). Figure 2 presents their distribution.

The five studies lacking Faithfulness comprise three human-centred evaluation designs [27, 76, 77] and two consistency-focused benchmarks [61, 71]—a non-random pattern addressed in the Discussion.

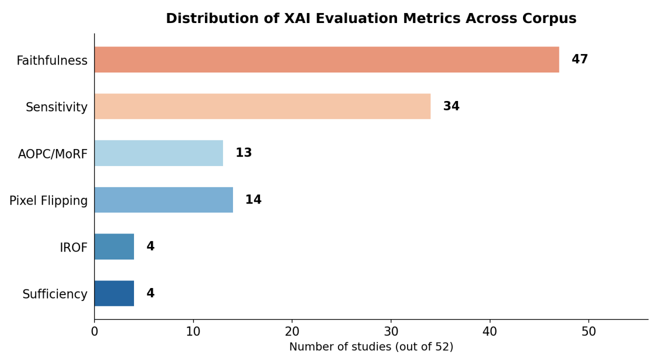

Figure 2: Distribution of XAI evaluation metrics across the 52-study corpus. Faithfulness is the most prevalent (47/52; 90.4%), while IROF and Sufficiency are each reported in only 4 studies (7.7%).

5.2 Metric Co-occurrence and Correlation

Pairwise co-occurrence analysis revealed three distinct patterns (Figures 4 and 3).

Pattern 1: Faithfulness–Sensitivity dominance. These co-occurred in 33 of the 47 Faithfulness-reporting studies (70.2%; 63.5% of the full corpus). Table 2 provides representative studies across domains and taxonomy layers, reflecting the cross-layer reach of these co-occurring metrics.

Pattern 2: Perturbation-based clustering. AOPC/MoRF and Pixel Flipping co-occurred in 7 studies, producing the strongest Spearman correlation ($\rho = 0.51$, $p < 0.001$). AOPC/MoRF and IROF also correlated significantly ($\rho = 0.37$, $p = 0.007$). The Faithfulness–Sensitivity correlation was modest but significant ($\rho = 0.33$, $p = 0.016$).

Pattern 3: Sufficiency isolation. Sufficiency exhibited negligible co-occurrence with all other metrics, producing negative correlations (ρ from -0.09 to -0.15, all non-significant at $\alpha = 0.05$).

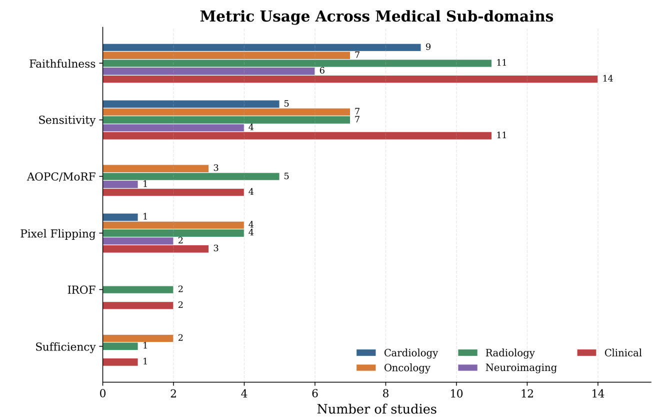

Figure 3: Metric usage across medical sub-domains. Each cell reports study count employing a given metric within a sub-domain.

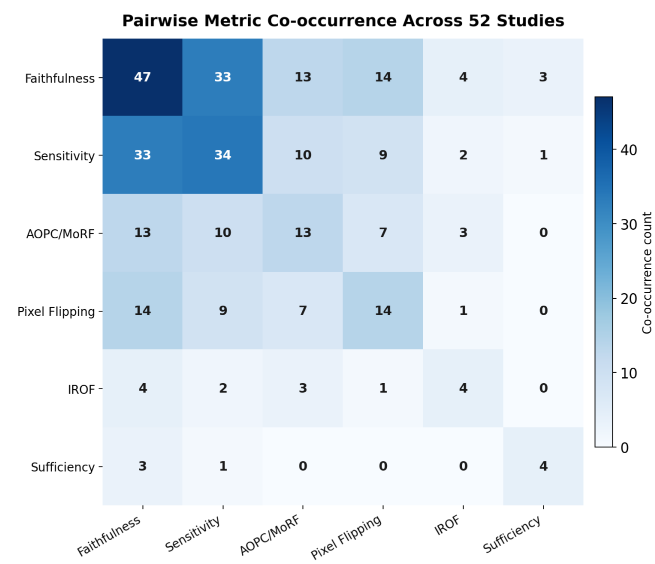

Figure 4: Pairwise metric co-occurrence across 52 studies. Diagonal entries indicate total frequency; off-diagonal entries indicate joint reporting.

[1]Studies categorised under Clinical AI, Clinical ML, Healthcare ML, and Medical Imaging are grouped under "Clinical" for cross-domain comparison.

Table 1: Evaluation studies included in the corpus ($N = 52$). Domains: Card. = Cardiology; Onc. = Oncology; Rad. = Radiology; Neuro. = Neuroimaging; Clin. = Clinical Decision Support.

Study	Domain	Venue	Notes
[54]	Card.	iScience	Perturbation-based saliency for 12-lead ECG
[37]	Card.	J. Clin. Med.	Grad-CAM sensitivity in prenatal echocardiography
[35]	Card.	Comput. Biol. Med.	SHAP inspection for heart-failure mortality
[36]	Card.	JMIR Med. Inform.	Interpretable HCM progression modelling
[38]	Card.	Front. Cardiovasc. Med.	Attribution stability for HF survival
[55]	Card.	FLAIRS	Conceptual explanation for ECG classification
[56]	Card.	Symmetry	Feature-level CHF explanation
[7]	Card.	Sci. Rep.	SHAP-based CVD risk prediction
[8]	Card.	Sci. Rep.	XAI chatbot for CVD screening
[42]	Onc.	Sci. Rep.	Saliency vs. lesion ground truth in breast cancer
[6]	Onc.	Sci. Rep.	Lesion-focused DL for breast cancer
[57]	Onc.	IEEE Access	SHAP/Grad-CAM bias analysis in breast cancer
[5]	Onc.	Healthcare	XAI meta-analysis for clinical decision support
[58]	Onc.	arXiv:2601.12826	Perturbation-based faithfulness + localization reliability for Grad-CAM in lung cancer CT classification
[59]	Onc.	Pattern Recognit.	Multi-task attention for COVID-19 CXR
[60]	Onc.	Eur. J. Radiol.	Quantitative saliency evaluation in mammography
[61]	Onc.	Int. J. Intell. Syst.	Fuzzy ensemble for cervical cancer
[40]	Rad.	Sci. Rep.	Heatmap localisation for cardiomegaly CXR
[62]	Rad.	J. Pers. Med.	Scalability evaluation for COVID-19 CXR
[63]	Rad.	IEEE TMI	Domain-aware interpretation; misleading attributions
[64]	Rad.	IEEE Access	Example-based XAI review in medical imaging
[10]	Rad.	Sensors	Bias-sensitive saliency for COVID CT
[41]	Rad.	Mach. Learn.	Robustness under corruption in medical images
[47]	Rad.	PLoS ONE	Human-centred Grad-CAM vs. LIME study
[65]	Rad.	Nat. Mach. Intell.	Saliency benchmarking for CXR
[66]	Rad.	Radiology: AI	Saliency trustworthiness for abnormality detection
[67]	Rad.	Radiology: AI	Attention saliency for pneumothorax
[68]	Rad.	Sci. Rep.	Patch perturbation for COVID-19 CXR
[69]	Rad.	BioData Mining	Visual explainability with statistical analysis
[45]	Neuro.	Sensors	Saliency sensitivity for brain tumour
[44]	Neuro.	Heliyon	XAI-augmented CNN for brain tumour
[70]	Neuro.	Med. Image Anal.	Counterfactual explanation evaluation
[46]	Neuro.	WIREs DMKD	XAI review in intracranial haemorrhage
[11]	Neuro.	MICCAI	Counterfactual impact analysis
[71]	Neuro.	SPIE ICBBE	Consistency benchmarking for brain MRI
[72]	Neuro.	Med. Eng. Phys.	LIME/Grad-CAM/SHAP for brain tumour
[73]	Clin.	IEEE J-BHI	Trustworthiness metrics for IoMT imaging
[48]	Clin.	IEEE Access	Co-design with radiologists for human-centred trust
[74]	Clin.	Artif. Intell. Med.	Explanation validation in CAD diagnosis
[39]	Clin.	IEEE PerCom Wksp.	qxAI faithfulness + sensitivity metrics
[75]	Clin.	Sci. Rep.	Domain-aware criteria beyond predictive metrics
[76]	Clin.	Int. J. HCI	Explainability effects on clinician trust
[77]	Clin.	KI-Künstl. Intell.	Human-centred lessons for medical decisions
[27]	Clin.	JAMIA	Human-centred explainability review
[78]	Clin.	Appl. Sci.	Grad-CAM usefulness across modalities
[2]	Clin.	Med. Image Anal.	Clinical evaluation guidelines
[79]	Clin.	WIREs Mech. Dis.	Attribution XAI survey in healthcare
[80]	Clin.	ACM Comput. Surv.	XAI survey with metrics benchmarking
[81]	Clin.	AAAI	Multi-modal XAI against clinical requirements
[43]	Clin.	Comput. Meth. Prog. Biomed.	Systematic faithfulness evaluation of DL methods
[82]	Clin.	Comput. Intell.	XAI experiments on infrared breast images
[83]	Clin.	Comput. Meth. Prog. Biomed.	Perturbation-based XAI for wound healing

Table 2: Representative studies by medical domain and taxonomy layer. Layer codes: L1 = Mathematical Faithfulness; L2 = System Robustness; L1/L2 = cross-layer (Sensitivity bridging both). Methods use canonical shorthand: attr. = attribution; bench. = benchmarking; CF = counterfactual; perturb. = perturbation.

Domain	Representative Studies	Layer	Methods
Cardiac	[35], [54], [38]	L1/L2	SHAP attr.; input perturb.; attr. stability
Oncology	[60], [42], [58]	L1/L2	AOPC; Max-Sensitivity; saliency–GT; concept faithfulness
Radiology	[65], [41], [66]	L1/L2	Saliency bench.; corruption robustness; trustworthiness
Brain/Neuro	[45], [71], [70]	L1/L2	Saliency sensitivity; consistency bench.; CF eval.
Clinical	[43], [81], [75]	L1/L2	Systematic Faith. eval.; multi-modal XAI; domain criteria

5.3 Cross-Domain Alignment

Faithfulness and Sensitivity qualified as near-cross-domain criteria, appearing in ≥ 4 sub-domains. Table 3 summarises representative Layer 2 (System Robustness) studies by domain; radiology contributes the largest cluster, consistent with the availability of saliency benchmarks under distributional shift.

IROF and Pixel Flipping were concentrated in radiology and oncology where saliency-map explanations are standard. The radiology sub-corpus exhibited the highest metric diversity; neuroimaging relied predominantly on Sensitivity with fewer metrics per study. Oncology studies were the most metrically diverse, combining technical and user-study-based criteria. The clinical sub-domain expanded most substantially, incorporating human-centred studies [27, 76, 77] that operationalise Layer 3 trust criteria.

5.4 Taxonomy Validation Against the Corpus

Mapping the 52 studies onto the three-layer taxonomy yielded: 47/52 (90.4%) reporting at least one Layer 1 criterion; 18 (34.6%) addressing Layer 2 robustness; and 7 (13.5%) including any Layer 3 human-centred measure.

Table 4 lists all seven Layer 3 studies exhaustively, confirming their concentration in the clinical sub-domain

Table 3: Representative Layer 2 (System Robustness) studies by domain. These 18 studies (34.6%) explicitly evaluate explanation stability under distributional shift, adversarial perturbation, or consistency degradation. Approach shorthand: nat. = natural; adv. = adversarial; perturb. = perturbation; bench. = benchmarking.

Domain	Representative Studies	Robustness Approach
Radiology	[41], [68], [10]	Corruption (nat. + adv.); patch perturb.; bias-sensitive saliency
Cardiac	[38], [54]	Attr. stability; input perturbation
Oncology	[43], [60], [58]	Max-Sensitivity; quantitative saliency; concept-based eval.
Brain/Neuro	[71], [45]	Consistency bench.; saliency sensitivity
Clinical	[75], [73]	Domain-aware criteria; IoMT trustworthiness

and their reliance on user- or clinician-facing evaluation designs. Cross-layer criteria—most notably contrastive explanations—appeared in several studies and were flagged accordingly.

Table 4: All Layer 3 (Human-Centred Trust) studies in the corpus ($n = 7$; 13.5%). Unlike Tables 2 and 3, this table is exhaustive rather than representative, reflecting the rarity of trust-layer evaluation. Type: U = clinician/user study; R = review or framework; G = clinical guideline.

Domain	Study	Trust Criterion	Type
Radiology	[47]	Grad-CAM vs. LIME; user preference	U
Clinical	[48]	Co-design with radiologists; trust alignment	U
Clinical	[76]	Clinician trust; task accuracy	U
Clinical	[27]	Human-centred eval. framework	R
Clinical	[77]	Decision quality; cognitive load	U
Clinical	[2]	Clinical evaluation guidelines	G
Clinical	[75]	Beyond-predictive trust criteria	R

6 Discussion

The results reveal both encouraging convergences and persistent gaps in medical XAI evaluation. This section interprets key findings, analyses metric redundancy, proposes a minimal evaluation framework, discusses clinical implications, and identifies threats to validity.

6.1 Interpretation of Key Findings

Expanding the corpus to 52 studies introduces an important revision to the Faithfulness finding. Rather than appearing universally, Faithfulness was present in 47/52 studies (90.4%), with the five exceptions concentrated in human-centred and consistency-focused designs [27,61,71,76,77]. This pattern is analytically significant: Faithfulness absence is not randomly distributed but co-occurs with a shift in evaluation layer, indicating a genuine conceptual divide between technical-fidelity and clinician-trust research paradigms. Studies prioritising clinician-facing evaluation operate under a distinct evaluative logic in which mathematical fidelity is not the primary concern—consistent with Rosenfeld's [19] observation that technical and human-centred evaluation priorities tend to diverge.

Because Faithfulness exhibits non-trivial variance, Spearman correlations involving it are computable. The Faithfulness–Sensitivity correlation ($\rho = 0.33$, $p = 0.016$) confirms co-occurrence without implying full redundancy: both metrics target Layer 1 but assess complementary properties—Faithfulness measures fidelity of attributed importance scores (Equation 1), while Sensitivity measures explanation stability under input perturbation (Equation 2).

6.2 Metric Redundancy and Complementarity

The strongest pairwise correlation is between AOPC/MoRF and Pixel Flipping ($\rho = 0.51$, $p < 0.001$), a result that is mechanistically expected: both are perturbationbased methods that degrade saliencyranked features and measure the resulting prediction drop. Critically, AOPC operates at the feature level (Equation 3), while Pixel Flipping operates at pixellevel granularity (Equation 4). Their high co-occurrence correlation reflects *methodological overlap*—both instantiate the same perturbationandmeasure paradigm—rather than strict *conceptual redundancy*. In practice, this means that reporting both metrics within a single study adds limited incremental information: selecting one perturbationbased metric (AOPC for featurelevel; Pixel Flipping for pixellevel) is typically sufficient.

AOPC/MoRF and IROF also correlate significantly ($\rho = 0.37$, $p = 0.007$). This pair exhibits a different relationship: AOPC removes individual features, whereas IROF removes semantically coherent segments (Equation 5). Their correlation thus reflects a shared perturbation paradigm, but IROF captures contributions of anatomically meaningful regions—making it complementary rather than redundant in medical imaging applications where anatomical structures are clinically relevant.

By contrast, Sufficiency exhibited negative correlations with all other metrics (ρ from -0.09 to -0.15, all non-significant). This isolation reflects a conceptual gap: Sufficiency asks whether the identified features *alone* suffice to reproduce the prediction (Equation 6), a question that the remaining metrics do not address. The negative correlations suggest that studies employing Sufficiency tend to adopt a fundamentally different evaluation philosophy, further supporting the position that Sufficiency captures a non-redundant evaluative dimension.

Figure 5 presents the full correlation matrix. Figure 6 visualises these relationships as a metric correlation network, highlighting the perturbation-based redundancy cluster and the isolation of Sufficiency.

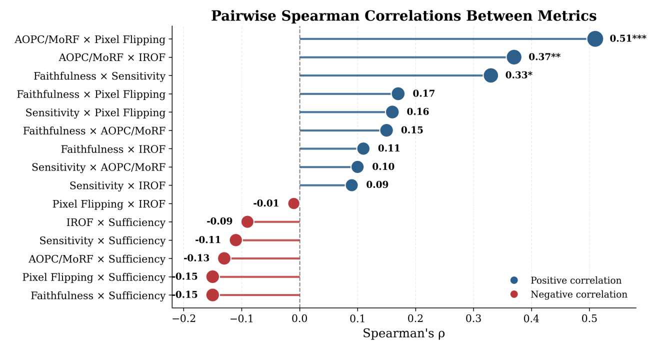

Figure 5: Spearman rank correlation (ρ) between XAI evaluation metrics across the 52-study corpus (equivalent to the phi coefficient for binary data). Statistical significance: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. The AOPC/MoRF–Pixel Flipping pair exhibits the strongest correlation ($\rho = 0.51$), while Sufficiency shows negative correlations with all other metrics.

The sub-domain disparity between oncology and neuroimaging further underscores that metric richness is unevenly distributed. Oncology benefits from well-established imaging benchmarks supporting diverse evaluation protocols, whereas neuroimaging lacks analogous shared resources, constraining both metric diversity and cross-study comparability. Targeted benchmark development for underserved sub-domains could have a disproportionately positive effect on standardisation.

6.3 Toward a Minimal Evaluation Framework

Building on the co-occurrence patterns, correlation analysis, and taxonomy mapping, we propose a minimal evaluation framework that spans all three taxonomy layers while imposing a tractable methodological burden. To our knowledge, this is the first such framework empirically grounded in cross-domain usage patterns rather than purely conceptual design. The framework specifies five evaluation dimensions, each addressed by at least one metric. Table 5 presents the framework in compact

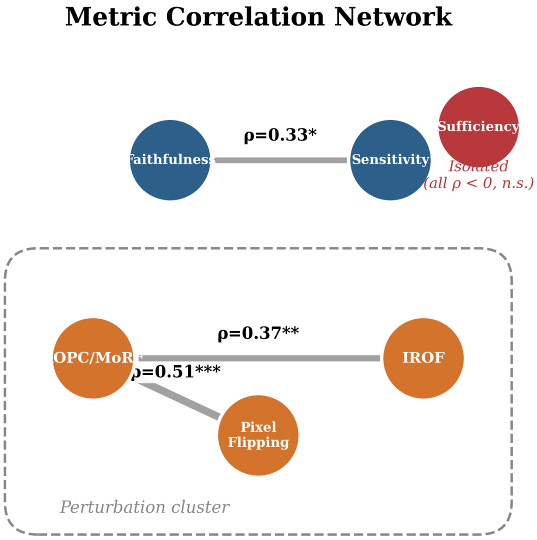

Figure 6: Metric correlation network derived from Spearman's ρ across the 52-study corpus. Nodes represent evaluation metrics; edge thickness and colour encode correlation strength. The dashed boundary highlights the perturbation-based redundancy cluster (AOPC/MoRF, Pixel Flipping, IROF). Sufficiency is visually isolated, confirming its non-redundant evaluative role. Only statistically significant correlations ($p < 0.05$) are shown as edges.

form; Figure 7 illustrates the three-layer structure.

This five-component framework ensures coverage of all three taxonomy layers without requiring simultaneous application of all six metrics, several of which exhibit partial redundancy. Researchers operating under resource constraints may prioritise Faithfulness, one perturbation metric, and one human-centred measure as a minimum viable set, provided that the omission of Sensitivity and Sufficiency is explicitly acknowledged and justified.

6.4 Clinical Implications

The structural imbalance documented by the taxonomy mapping carries direct clinical consequences. Two omission risks are particularly concerning.

Risk of missing Sufficiency. When Sufficiency is not evaluated, there is no guarantee that the features highlighted by an explanation are, on their own, adequate to sustain the model's prediction. In clinical practice, this means a clinician may act on an explanation that identifies *correlated* but not *causally sufficient* features—potentially leading to diagnostic decisions based on incomplete evidence. For example, a saliency map highlighting peri-lesion tissue rather than the lesion itself may achieve high Faithfulness scores (the model *does* attend to that region) while failing Sufficiency (the highlighted

Table 5: Proposed minimal evaluation framework for medical XAI. Each row specifies one required evaluation dimension, its taxonomy layer, evaluative role, and rationale. † indicates that the metric may be omitted under resource constraints if the omission is explicitly justified.

Metric	Layer	Req.	Rationale
Faithfulness	L1	Req.	Technical credibility foundation; 90.4% adoption
Pert. (AOPC/PF/IROF)	L1	Req.	Validates removal impact; one sufficient ($\rho = 0.51$)
Sensitivity	L1/L2	Req.†	Links faithfulness–robustness; input responsiveness
Sufficiency	L1	Req.†	Assesses decision-sufficient evidence; 7.7% adoption
Human-centred	L3	Req.	Essential for deployment; 13.5% coverage

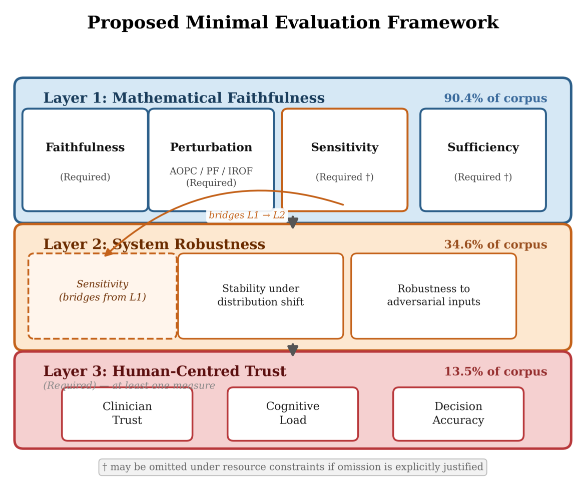

Figure 7: Proposed minimal evaluation framework organised by taxonomy layer. Layer 1 (Mathematical Faithfulness) requires Faithfulness, one perturbation metric, Sensitivity, and Sufficiency. Layer 2 (System Robustness) is partially addressed by Sensitivity. Layer 3 (Human-Centred Trust) requires at least one clinician-facing measure. Arrows indicate cross-layer bridging by Sensitivity.

region alone does not reproduce the prediction). Without Sufficiency evaluation, such failure modes remain undetected.

Risk of missing Robustness. When robustness is not evaluated (absent in 65.4% of corpus studies), explanations may be valid only under the specific data distribution on which they were generated. In multi-site clinical deployment—where imaging equipment, patient demographics, and acquisition protocols vary across hospitals—an explanation that is faithful under training conditions may become unstable or misleading under distributional shift. Repetto et al. [41] provide empirical evidence that XAI explanations degrade meaningfully under both natural and adversarial corruption, confirming that robustness cannot be assumed but must be explicitly tested prior to clinical deployment.

6.5 Limitations

This study has several limitations. First, metric classification into canonical categories requires interpretive judgement. Although we followed established normalisation procedures [19, 20], different coding decisions could alter metric distribution. Binary encoding further simplifies reporting by treating all mentions equally, regardless of depth or emphasis. While the screening process involved two independent reviewers with substantial agreement (Cohen's $\kappa = 0.82$), some degree of selection bias cannot be excluded.

Second, the corpus of 52 studies spans five sub-domains but remains modest in size, limiting statistical power for low-frequency metrics such as IROF and Sufficiency (each $n = 4$). Observed patterns should therefore be interpreted as indicative rather than definitive, and correlations involving rare metrics require caution. Additionally, the restriction to medical XAI limits generalisability to other high-stakes domains, although prior work suggests similar fragmentation elsewhere [84, 85].

Third, co-occurrence analysis captures which metrics are reported together but cannot establish true redundancy or agreement in evaluation outcomes. Spearman correlations reflect associations in reporting patterns rather than equivalence in measurement. Establishing redundancy would require controlled comparisons on shared explanation–dataset pairs, which is beyond the scope of this review.

Finally, the proposed three-layer taxonomy is primarily derived from deductive synthesis and has not been validated with clinicians. Prospective user studies are needed to confirm whether the trust layer reflects real-world clinical requirements. In addition, the focus on English-language publications in indexed venues may underrepresent practices from other research communities.

6.6 Future Work

Three directions follow from these findings. First, a prospective study instrumenting clinicians as they interact with SHAP- and Grad-CAM-generated explanations would ground the trust layer in behavioural evidence. Second, developing a standardised reporting template—specifying the minimal metric set proposed in Section 6.3 alongside optional domain-specific additions—could reduce the documented fragmentation. Third, extending analyses to include user-study metrics such as cognitive load and decision accuracy would enable testing whether high Faithfulness scores translate into genuine clinical utility.

7 Conclusion

This paper investigated the evaluation landscape of XAI within medical AI through a systematically assembled corpus of 52 peer-reviewed studies spanning five sub-domains. To our knowledge, this constitutes the first cross-domain co-occurrence analysis of XAI evaluation metrics in medical AI. We identified a dominant cluster of technically grounded metrics—Faithfulness and Sensitivity—that consistently co-occur. Perturbation-based metrics form a correlated secondary cluster ($\rho = 0.51$, $p < 0.001$), exhibiting partial methodological redundancy, while Sufficiency remains isolated (7.7%) despite direct relevance to clinical decision adequacy.

Mapping the corpus onto a three-layer taxonomy confirmed that evaluation is heavily concentrated in mathematical faithfulness (90.4%), with system robustness (34.6%) and human-centred trust (13.5%) substantially underrepresented—leaving the dimensions most consequential for clinical deployment systematically underserved.

These findings support two conclusions. First, a minimal evaluation framework—combining one faithfulness measure, one perturbation-based measure, one sensitivity metric, one sufficiency check, and at least one human-centred criterion—covers all three taxonomy layers without unreasonable methodological burden. Second, the field requires shared benchmarks and reporting standards, particularly for underserved sub-domains such as neuroimaging. Addressing these gaps is necessary for medical AI systems whose explanations are not only technically faithful but genuinely trustworthy in clinical practice.

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